THERMAL ENGINEERING LAB

III B. Tech. I Semester: MECHANICAL ENGINEERING								
Course Code	**Category**	**Hours / Week**			**Credits**	**Maximum Marks**		
A6ME28	**PCC**	**L**	**T**	**P**	**C**	**CIE**	**SEE**	**Total**
		0	0	2	1	40	60	100

COURSE OVERVIEW:

The primary purpose of the 'Thermal Engineering Lab' is to show students the experimental methods on thermal energies on various engines and demonstrate their operational procedures. These values can further be used to determine other fuel properties. In order that students have good understanding of the theory underlying the experiments, the entire course is designed such that classroom lectures precede lab work. Students are advised to pay close attention in class so that they can perform well in the lab.

COURSE OUTCOMES:

At the end of the course, the student should be able to:

1. Illustrate the Valve timing and port timing diagrams of CI & SI Engines.
2. Examine the performance parameters of internal combustion engines.
3. Examine the heat balance sheet in an internal combustion engine
4. Plot the performance characteristics of reciprocating air compressor.
5. Understand the working of steam boilers.

LIST OF EXPERIMENTS:

1. I.C. Engines Valve Timing Diagram
2. I.C. Engines Port Timing Diagram
3. I.C. Engines Performance Test 4-Stroke Diesel Engine
4. I.C. Engines Performance Test 4-Stroke Petrol Engine
5. I.C. Engines Heat Balance
6. I.C. Engines Air/Fuel Ratio and Volumetric Efficiency
7. Performance Test on Variable Compression Ratio Engine
8. Performance Test on Reciprocating Air–Compressor Unit
9. Dis-assembly/ Assembly of Engines
10. Study of Boilers